

- 15** In a titration experiment,  $25.0 \text{ cm}^3$  of a solution of a weak acid, HA, required  $V \text{ cm}^3$  of NaOH solution to neutralise completely.

In a separate titration,  $\frac{1}{2}V \text{ cm}^3$  of the same NaOH solution was added to  $25.0 \text{ cm}^3$  of the same solution of HA. The pH was measured at this point.

How are the value of  $V$  and the pH at  $\frac{1}{2}V$  affected when the experiment was repeated at a higher temperature?

|          | value of $V$     | pH at $\frac{1}{2}V$ |
|----------|------------------|----------------------|
| <b>A</b> | decreases        | decreases            |
| <b>B</b> | decreases        | remains the same     |
| <b>C</b> | remains the same | decreases            |
| <b>D</b> | remains the same | remains the same     |